

LOSO (glove_ml_explorer) Experiment SB_flex

Static · flex only (both hands), no Nothing

Generated 2026-05-12 00:06:49

Experiment configuration

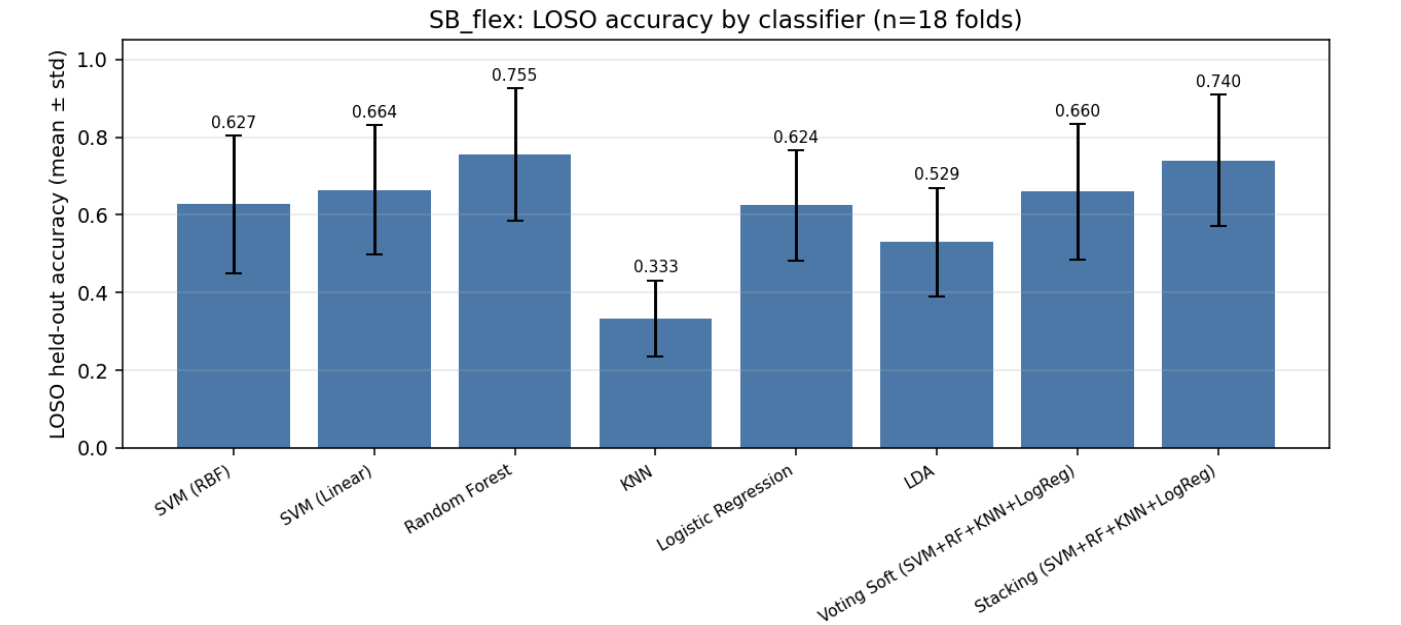
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	flex
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	20
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	18: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn, Stephen, Val

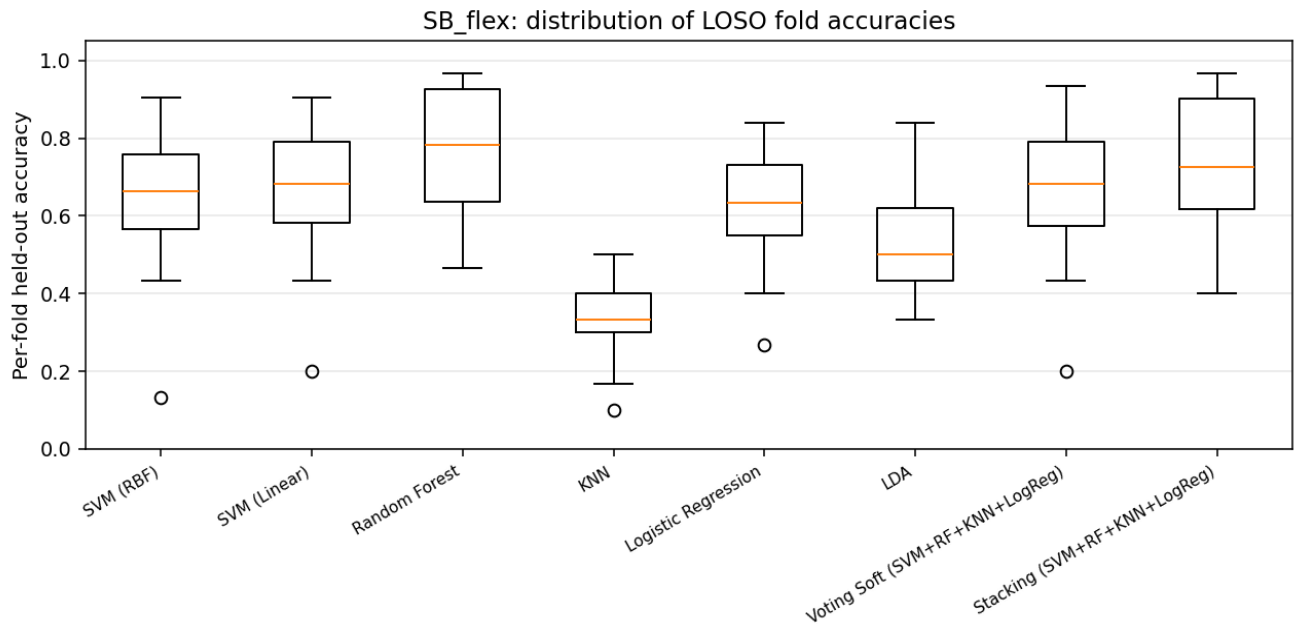
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.6272	0.1771	0.1333	0.9032	0.7606
SVM (Linear)	16,215	0.6645	0.1670	0.2000	0.9032	0.7522
Random Forest	30,376	0.7553	0.1695	0.4667	0.9677	0.9288
KNN	—	0.3335	0.0982	0.1000	0.5000	0.4690
Logistic Regression	6,486	0.6243	0.1420	0.2667	0.8387	0.7207
LDA	6,486	0.5294	0.1403	0.3333	0.8387	0.5774
Voting Soft (SVM+RF+KNN+LogReg)	—	0.6595	0.1745	0.2000	0.9355	0.7980
Stacking (SVM+RF+KNN+LogReg)	—	0.7396	0.1694	0.4000	0.9677	0.8852

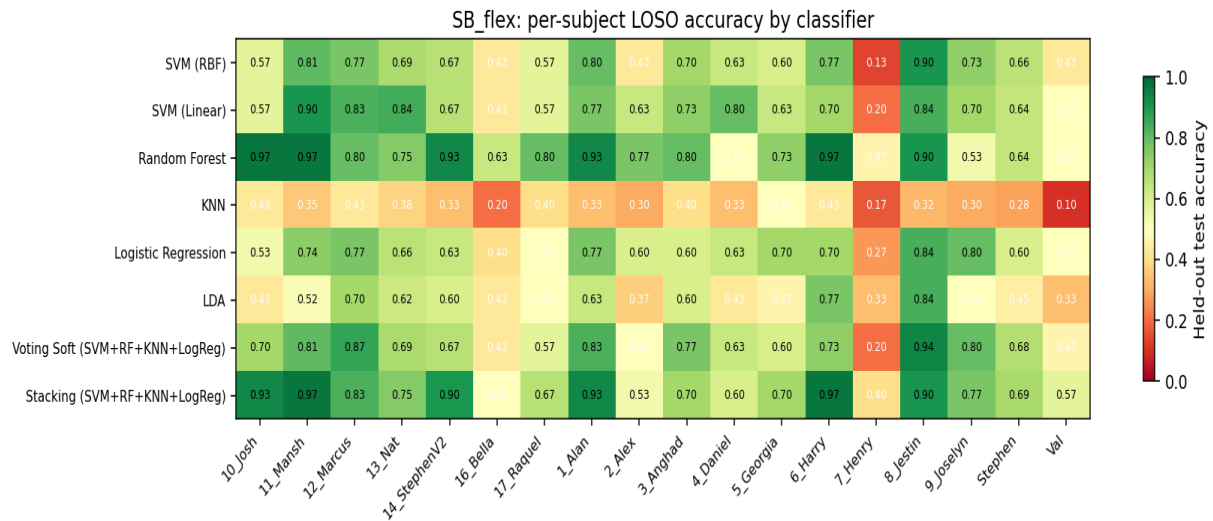
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	0.567	0.567	0.967	0.433	0.533	0.433	0.700	0.933
11_Mansh	603	0.806	0.903	0.968	0.355	0.742	0.516	0.806	0.968
12_Marcus	604	0.767	0.833	0.800	0.433	0.767	0.700	0.867	0.833
13_Nat	602	0.688	0.844	0.750	0.375	0.656	0.625	0.688	0.750
14_StephenV2	604	0.667	0.667	0.933	0.333	0.633	0.600	0.667	0.900
16_Bella	604	0.433	0.433	0.633	0.200	0.400	0.433	0.433	0.500
17_Raquel	604	0.567	0.567	0.800	0.400	0.500	0.500	0.567	0.667
1_Alan	604	0.800	0.767	0.933	0.333	0.767	0.633	0.833	0.933
2_Alex	604	0.433	0.633	0.767	0.300	0.600	0.367	0.500	0.533
3_Anghad	604	0.700	0.733	0.800	0.400	0.600	0.600	0.767	0.700
4_Daniel	604	0.633	0.800	0.500	0.333	0.633	0.433	0.633	0.600
5_Georgia	604	0.600	0.633	0.733	0.500	0.700	0.467	0.600	0.700
6_Harry	604	0.767	0.700	0.967	0.433	0.700	0.767	0.733	0.967
7_Henry	604	0.133	0.200	0.467	0.167	0.267	0.333	0.200	0.400
8_Jestin	603	0.903	0.839	0.903	0.323	0.839	0.839	0.935	0.903
9_Joselyn	604	0.733	0.700	0.533	0.300	0.800	0.500	0.800	0.767
Stephen	514	0.658	0.642	0.642	0.283	0.600	0.450	0.675	0.692
Val	604	0.433	0.500	0.500	0.100	0.500	0.333	0.467	0.567

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.6272 ± 0.1771 (min 0.1333, max 0.9032)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	4/5	4/5	2/5	3/5	1/5	0.567
11_Mansh	6/6	5/5	4/5	4/5	5/5	1/5	0.806
12_Marcus	5/5	4/5	2/5	4/5	5/5	3/5	0.767
13_Nat	5/6	5/5	5/5	3/5	1/6	3/5	0.688
14_StephenV2	5/5	2/5	5/5	3/5	0/5	5/5	0.667
16_Bella	5/5	4/5	0/5	4/5	0/5	0/5	0.433
17_Raquel	2/5	0/5	5/5	1/5	4/5	5/5	0.567
1_Alan	3/5	4/5	5/5	4/5	3/5	5/5	0.800
2_Alex	1/5	4/5	2/5	2/5	2/5	2/5	0.433
3_Anghad	4/5	5/5	5/5	1/5	5/5	1/5	0.700
4_Daniel	4/5	5/5	3/5	3/5	1/5	3/5	0.633
5_Georgia	3/5	5/5	0/5	5/5	3/5	2/5	0.600
6_Harry	4/5	5/5	3/5	4/5	4/5	3/5	0.767
7_Henry	3/5	0/5	0/5	0/5	1/5	0/5	0.133
8_Jestin	6/6	3/5	5/5	5/5	4/5	5/5	0.903
9_Joselyn	5/5	2/5	4/5	5/5	4/5	2/5	0.733
Stephen	5/20	11/20	19/20	18/20	16/20	10/20	0.658
Val	1/5	2/5	5/5	0/5	4/5	1/5	0.433
MEAN	0.705	0.686	0.686	0.606	0.587	0.494	0.627

Classifier: SVM (Linear)

Params (best-effort): 16,215

LOSO mean accuracy: 0.6645 ± 0.1670 (min 0.2000, max 0.9032)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	4/5	4/5	2/5	2/5	2/5	0.567
11_Mansh	6/6	5/5	4/5	5/5	5/5	3/5	0.903
12_Marcus	5/5	5/5	2/5	4/5	5/5	4/5	0.833
13_Nat	6/6	5/5	5/5	3/5	4/6	4/5	0.844
14_StephenV2	5/5	4/5	5/5	1/5	0/5	5/5	0.667
16_Bella	5/5	4/5	0/5	4/5	0/5	0/5	0.433
17_Raquel	1/5	0/5	5/5	2/5	4/5	5/5	0.567
1_Alan	3/5	4/5	4/5	4/5	3/5	5/5	0.767
2_Alex	4/5	4/5	4/5	0/5	2/5	5/5	0.633
3_Anghad	4/5	5/5	5/5	1/5	5/5	2/5	0.733
4_Daniel	5/5	5/5	3/5	5/5	4/5	2/5	0.800
5_Georgia	4/5	5/5	1/5	5/5	3/5	1/5	0.633
6_Harry	5/5	5/5	3/5	4/5	2/5	2/5	0.700
7_Henry	4/5	0/5	1/5	0/5	1/5	0/5	0.200
8_Jestin	6/6	3/5	3/5	5/5	4/5	5/5	0.839
9_Joselyn	5/5	0/5	5/5	4/5	5/5	2/5	0.700
Stephen	2/20	7/20	19/20	19/20	17/20	13/20	0.642
Val	1/5	3/5	5/5	0/5	5/5	1/5	0.500
MEAN	0.772	0.697	0.708	0.597	0.640	0.569	0.664

Classifier: Random Forest

Params (best-effort): 30,376

LOSO mean accuracy: 0.7553 ± 0.1695 (min 0.4667, max 0.9677)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	4/5	5/5	5/5	5/5	0.967
11_Mansh	6/6	4/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	0/5	5/5	5/5	4/5	0.800
13_Nat	6/6	5/5	5/5	5/5	0/6	3/5	0.750
14_StephenV2	5/5	5/5	5/5	3/5	5/5	5/5	0.933
16_Bella	5/5	4/5	0/5	5/5	5/5	0/5	0.633
17_Raquel	4/5	0/5	5/5	5/5	5/5	5/5	0.800
1_Alan	5/5	3/5	5/5	5/5	5/5	5/5	0.933
2_Alex	3/5	5/5	3/5	2/5	5/5	5/5	0.767
3_Anghad	2/5	5/5	5/5	4/5	3/5	5/5	0.800
4_Daniel	5/5	5/5	0/5	5/5	0/5	0/5	0.500
5_Georgia	5/5	5/5	4/5	5/5	1/5	2/5	0.733
6_Harry	5/5	5/5	4/5	5/5	5/5	5/5	0.967
7_Henry	5/5	2/5	0/5	2/5	1/5	4/5	0.467
8_Jestin	6/6	2/5	5/5	5/5	5/5	5/5	0.903
9_Joselyn	5/5	4/5	1/5	5/5	1/5	0/5	0.533
Stephen	8/20	3/20	20/20	14/20	20/20	12/20	0.642
Val	4/5	5/5	4/5	0/5	2/5	0/5	0.500
MEAN	0.889	0.775	0.667	0.828	0.700	0.678	0.755

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.3335 ± 0.0982 (min 0.1000, max 0.5000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	4/5	3/5	2/5	0/5	0/5	0.433
11_Mansh	3/6	4/5	0/5	3/5	1/5	0/5	0.355
12_Marcus	3/5	3/5	0/5	1/5	3/5	3/5	0.433
13_Nat	3/6	3/5	4/5	2/5	0/6	0/5	0.375
14_StephenV2	3/5	1/5	3/5	3/5	0/5	0/5	0.333
16_Bella	3/5	0/5	2/5	0/5	0/5	1/5	0.200
17_Raquel	1/5	0/5	5/5	4/5	2/5	0/5	0.400
1_Alan	1/5	1/5	3/5	2/5	2/5	1/5	0.333
2_Alex	0/5	4/5	2/5	3/5	0/5	0/5	0.300
3_Anghad	2/5	3/5	1/5	1/5	3/5	2/5	0.400
4_Daniel	4/5	2/5	0/5	2/5	1/5	1/5	0.333
5_Georgia	2/5	5/5	0/5	4/5	2/5	2/5	0.500
6_Harry	4/5	4/5	1/5	3/5	1/5	0/5	0.433
7_Henry	1/5	0/5	1/5	0/5	2/5	1/5	0.167
8_Jestin	0/6	4/5	0/5	2/5	3/5	1/5	0.323
9_Joselyn	0/5	1/5	0/5	2/5	3/5	3/5	0.300
Stephen	7/20	15/20	9/20	1/20	0/20	2/20	0.283
Val	0/5	2/5	0/5	1/5	0/5	0/5	0.100
MEAN	0.386	0.497	0.303	0.392	0.256	0.172	0.333

Classifier: Logistic Regression

Params (best-effort): 6,486

LOSO mean accuracy: 0.6243 ± 0.1420 (min 0.2667, max 0.8387)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	3/5	4/5	1/5	2/5	2/5	0.533
11_Mansh	5/6	5/5	3/5	4/5	5/5	1/5	0.742
12_Marcus	5/5	3/5	2/5	4/5	5/5	4/5	0.767
13_Nat	3/6	5/5	5/5	2/5	4/6	2/5	0.656
14_StephenV2	5/5	3/5	5/5	1/5	0/5	5/5	0.633
16_Bella	5/5	3/5	0/5	4/5	0/5	0/5	0.400
17_Raquel	0/5	0/5	5/5	1/5	4/5	5/5	0.500
1_Alan	3/5	4/5	5/5	3/5	3/5	5/5	0.767
2_Alex	3/5	3/5	4/5	2/5	2/5	4/5	0.600
3_Anghad	3/5	4/5	5/5	0/5	5/5	1/5	0.600
4_Daniel	5/5	5/5	2/5	4/5	0/5	3/5	0.633
5_Georgia	3/5	5/5	3/5	5/5	3/5	2/5	0.700
6_Harry	4/5	5/5	3/5	4/5	1/5	4/5	0.700
7_Henry	5/5	1/5	1/5	0/5	0/5	1/5	0.267
8_Jestin	6/6	3/5	4/5	5/5	3/5	5/5	0.839
9_Joselyn	5/5	2/5	4/5	5/5	5/5	3/5	0.800
Stephen	2/20	5/20	18/20	19/20	15/20	13/20	0.600
Val	1/5	3/5	5/5	0/5	5/5	1/5	0.500
MEAN	0.702	0.647	0.717	0.553	0.556	0.569	0.624

Classifier: LDA

Params (best-effort): 6,486

LOSO mean accuracy: 0.5294 ± 0.1403 (min 0.3333, max 0.8387)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	2/5	1/5	4/5	3/5	2/5	1/5	0.433
11_Mansh	6/6	2/5	1/5	3/5	3/5	1/5	0.516
12_Marcus	4/5	3/5	4/5	3/5	3/5	4/5	0.700
13_Nat	4/6	4/5	4/5	3/5	3/6	2/5	0.625
14_StephenV2	5/5	2/5	3/5	1/5	2/5	5/5	0.600
16_Bella	5/5	3/5	0/5	4/5	1/5	0/5	0.433
17_Raquel	1/5	2/5	4/5	1/5	2/5	5/5	0.500
1_Alan	3/5	4/5	3/5	4/5	1/5	4/5	0.633
2_Alex	1/5	0/5	4/5	2/5	3/5	1/5	0.367
3_Anghad	2/5	3/5	4/5	1/5	4/5	4/5	0.600
4_Daniel	1/5	4/5	3/5	4/5	0/5	1/5	0.433
5_Georgia	1/5	4/5	2/5	5/5	1/5	1/5	0.467
6_Harry	5/5	3/5	5/5	3/5	4/5	3/5	0.767
7_Henry	1/5	3/5	0/5	0/5	2/5	4/5	0.333
8_Jestin	4/6	5/5	3/5	5/5	4/5	5/5	0.839
9_Joselyn	3/5	1/5	0/5	4/5	4/5	3/5	0.500
Stephen	2/20	1/20	19/20	17/20	10/20	5/20	0.450
Val	2/5	3/5	3/5	1/5	1/5	0/5	0.333
MEAN	0.535	0.525	0.575	0.569	0.467	0.503	0.529

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.6595 ± 0.1745 (min 0.2000, max 0.9355)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	4/5	4/5	4/5	4/5	2/5	0.700
11_Mansh	6/6	5/5	3/5	4/5	5/5	2/5	0.806
12_Marcus	5/5	5/5	2/5	4/5	5/5	5/5	0.867
13_Nat	6/6	5/5	5/5	3/5	2/6	1/5	0.688
14_StephenV2	5/5	2/5	5/5	3/5	0/5	5/5	0.667
16_Bella	5/5	4/5	0/5	4/5	0/5	0/5	0.433
17_Raquel	1/5	0/5	5/5	2/5	5/5	4/5	0.567
1_Alan	3/5	4/5	5/5	4/5	4/5	5/5	0.833
2_Alex	1/5	4/5	3/5	2/5	2/5	3/5	0.500
3_Anghad	4/5	5/5	5/5	1/5	5/5	3/5	0.767
4_Daniel	5/5	5/5	3/5	4/5	0/5	2/5	0.633
5_Georgia	4/5	5/5	0/5	5/5	2/5	2/5	0.600
6_Harry	5/5	5/5	3/5	5/5	1/5	3/5	0.733
7_Henry	4/5	0/5	0/5	0/5	1/5	1/5	0.200
8_Jestin	6/6	3/5	5/5	5/5	5/5	5/5	0.935
9_Joselyn	5/5	2/5	4/5	5/5	4/5	4/5	0.800
Stephen	4/20	10/20	18/20	19/20	16/20	14/20	0.675
Val	1/5	4/5	5/5	0/5	3/5	1/5	0.467
MEAN	0.744	0.717	0.683	0.664	0.574	0.572	0.660

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.7396 ± 0.1694 (min 0.4000, max 0.9677)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	5/5	5/5	5/5	5/5	5/5	0.933
11_Mansh	6/6	5/5	5/5	5/5	5/5	4/5	0.968
12_Marcus	5/5	5/5	2/5	4/5	5/5	4/5	0.833
13_Nat	6/6	5/5	5/5	4/5	1/6	3/5	0.750
14_StephenV2	5/5	5/5	5/5	3/5	4/5	5/5	0.900
16_Bella	5/5	5/5	0/5	5/5	0/5	0/5	0.500
17_Raquel	4/5	0/5	5/5	1/5	5/5	5/5	0.667
1_Alan	5/5	4/5	5/5	5/5	4/5	5/5	0.933
2_Alex	3/5	4/5	2/5	2/5	2/5	3/5	0.533
3_Anghad	2/5	3/5	5/5	2/5	4/5	5/5	0.700
4_Daniel	5/5	5/5	0/5	5/5	1/5	2/5	0.600
5_Georgia	5/5	4/5	3/5	5/5	2/5	2/5	0.700
6_Harry	5/5	5/5	4/5	5/5	5/5	5/5	0.967
7_Henry	5/5	0/5	0/5	2/5	1/5	4/5	0.400
8_Jestin	6/6	2/5	5/5	5/5	5/5	5/5	0.903
9_Joselyn	5/5	2/5	4/5	5/5	4/5	3/5	0.767
Stephen	6/20	3/20	20/20	19/20	19/20	16/20	0.692
Val	2/5	5/5	5/5	0/5	4/5	1/5	0.567
MEAN	0.839	0.719	0.722	0.753	0.684	0.722	0.740